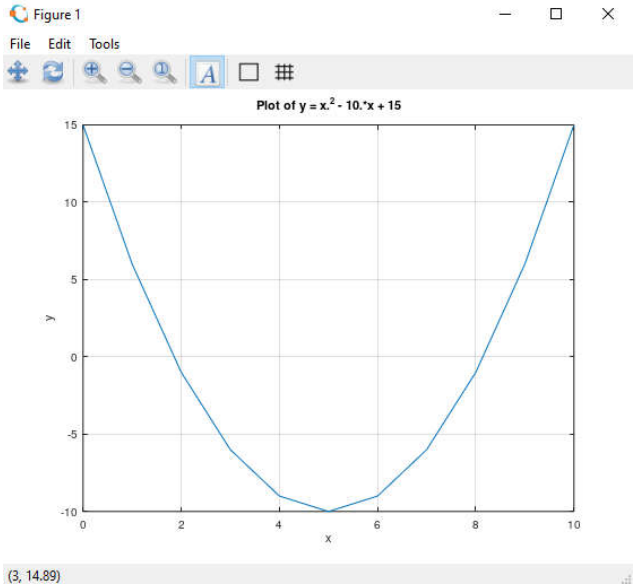
INTRODUCTION TO PLOTTING

The "plot" function

```
>> x = 0:1:10
>> y = x.^2 - 10.*x + 15
>> plot(x,y)
>> title ('Plot of y = x.^2 - 10.*x + 15')
>> xlabel ('x')
>> ylabel ('y')
>> grid on
```



- To save a plot, go to file, and save as or print the plot;
  Syntax: `print options filename`

E.g. to save a png file of 300 dpi resolution,

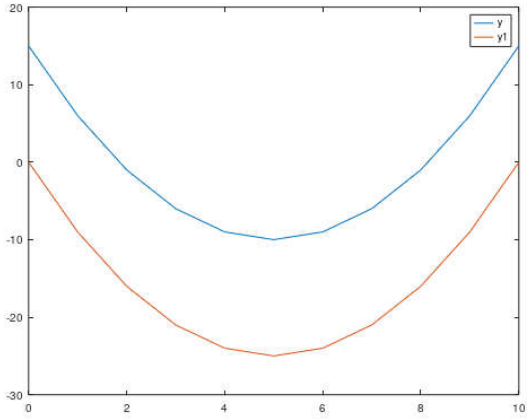
```
>> print -dpng -r300 my_image.png
```

Other filetype options include djpeg, dtiff, etc.

The "plot" function

Multiple plots on one sheet

```
>> x = 0:1:10
>> y1 = x.^2 - 10.*x
>> plot(x,y, x,y1)
```



- To add legends

```
>> legend('y','y1','location','northeast')
```

- Note: Location and northeast are not case sensitive in matlab. And it can also accept "NE"

```
>> legend('y','y1','Location',NE')
```


The "plot" function

- To plot with different colors and markers

```
>> plot(x,y, 'b--bo', x,y1,'r:r^')
```

line dashed blue markers blue circles

Line red dotted marker red triangle up

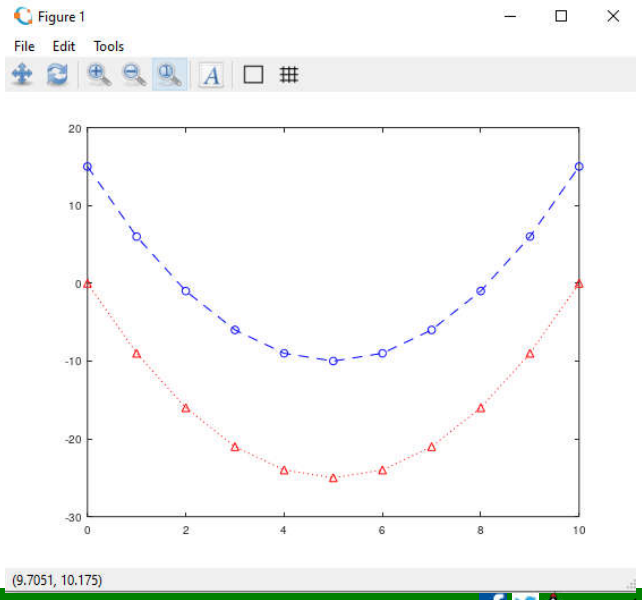
```
>>  plot(x,y,'b--',x,y,'bo',x,y1,'r:',x,y1,'r^')
```

Color		Marker Style		Line Style	
y	yellow	.	point	-	solid
m	magenta	o	circle	:	dotted
c	cyan	x	x-mark	-.	dash-dot
r	red	+	plus	--	dashed
g	green	*	star	<none>	no line
b	blue	s	square		
w	white	d	diamond		
k	black	v	triangle (down)		
		^	triangle (up)		
		<	triangle (left)		
		>	triangle (right)		
		p	pentagram		
		h	hexagram		
		<none>	no marker		

The "plot" function

- To plot with different colors and markers

```
>> plot(x,y,'b--o', x,y1,'r:^')
```

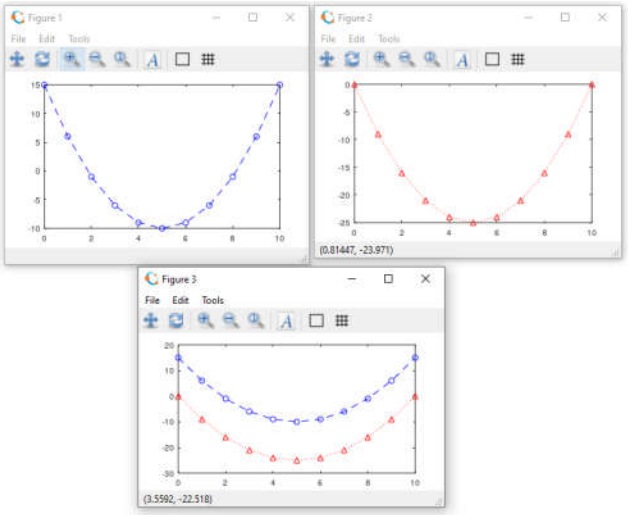


Multiple plots

- Multiple plots

- Use figure command to create multiple plots in different windows

```
>> plot(x,y,'b--o')
>> figure(2)
>> plot(x,y1,'r:^')
>> figure(3)
>> plot(x,y,'b--o', x,y1,'r:^')
```



Multiple plots

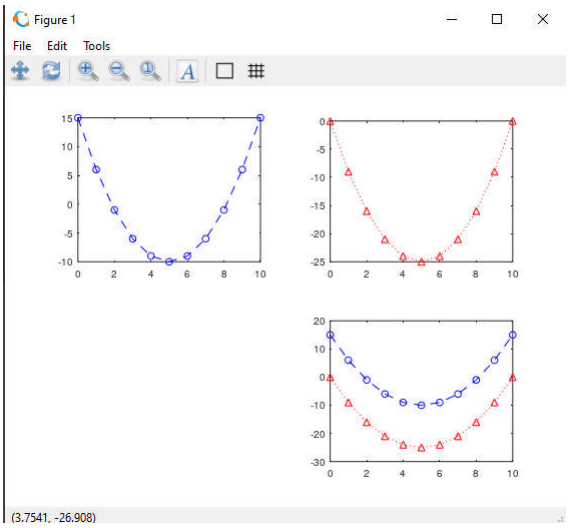
- Multiple plots on one Figure
- Use "subplot" function to divide one figure into rectangular panels that are numbered row-wise and plot in a specified panel.
- Syntax;

```
>>    subplot(m, n, p)
```

  - m=number of rows, n = number of columns, p=focus or replace

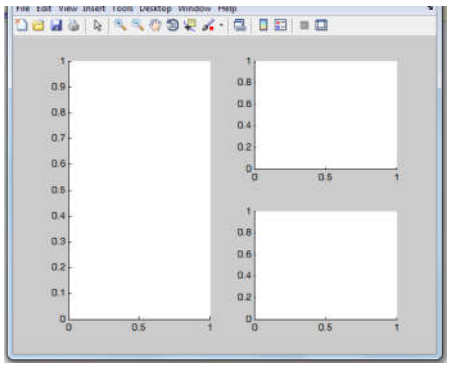
E.g.

```
>> subplot(2,2,1); plot(x,y,'b--o')
>> subplot(2,2,2)
>> plot(x,y1,'r:^')
>> subplot(2,2,4); plot(x,y,'b--o', x,y1,'r:^')
```

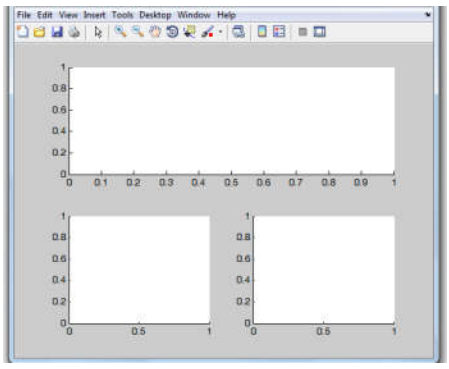


Multiple plots

```
>>    subplot(2,2,[1 3]); subplot (2,2,2); subplot (2,2,4)
```



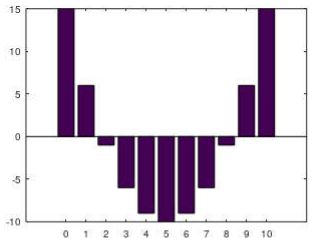
```
>>    subplot(2,2,[1 2]); subplot (2,2,3); subplot (2,2,4)
```

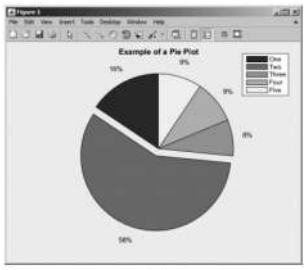


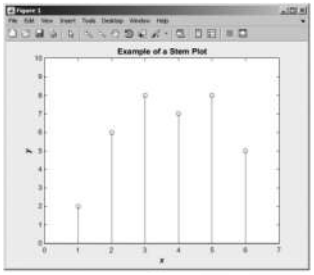
Other types of plots

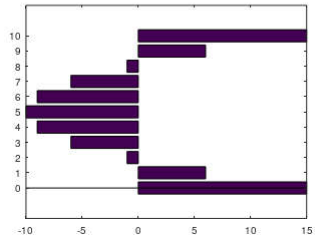
`bar(x,y)`	This function creates a *vertical* bar plot, with the values in `x` used to label each bar and the values in `y` used to determine the height of the bar.
`barh(x,y)`	This function creates a *horizontal* bar plot, with the values in `x` used to label each bar and the values in `y` used to determine the horizontal length of the bar.
`compass(x,y)`	This function creates a polar plot, with an arrow drawn from the origin to the location of each (*x, y*) point. Note that the locations of the points to plot are specified in Cartesian coordinates, not polar coordinates.
`pie(x)` `pie(x,explode)`	This function creates a pie plot. This function determines the percentage of the total pie corresponding to each value of `x`, and plots pie slices of that size. The optional array `explode` controls whether or not individual pie slices are separated from the remainder of the pie.
`stairs(x,y)`	This function creates a stair plot, with each stair step centered on an (*x, y*) point.
`stem(x,y)`	This function creates a stem plot, with a marker at each *(x, y)* point and a stem drawn vertically from that point to the *x*-axis.

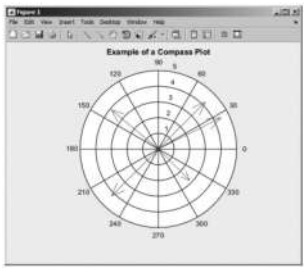
Other types of plots

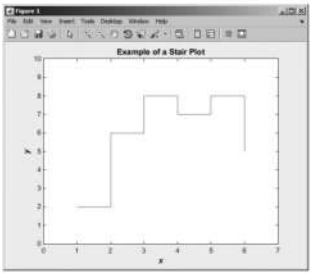












Other types of plots

```
>>  t = 0:pi/50:10*pi;
>>  plot3(sin(t),cos(t),t)
>>  grid on
>>  axis square
```

